

Adaptive Instance-Level Reweighting for Imbalanced Classification:

A Comparative Study on Credit Card Fraud and Mammography Datasets

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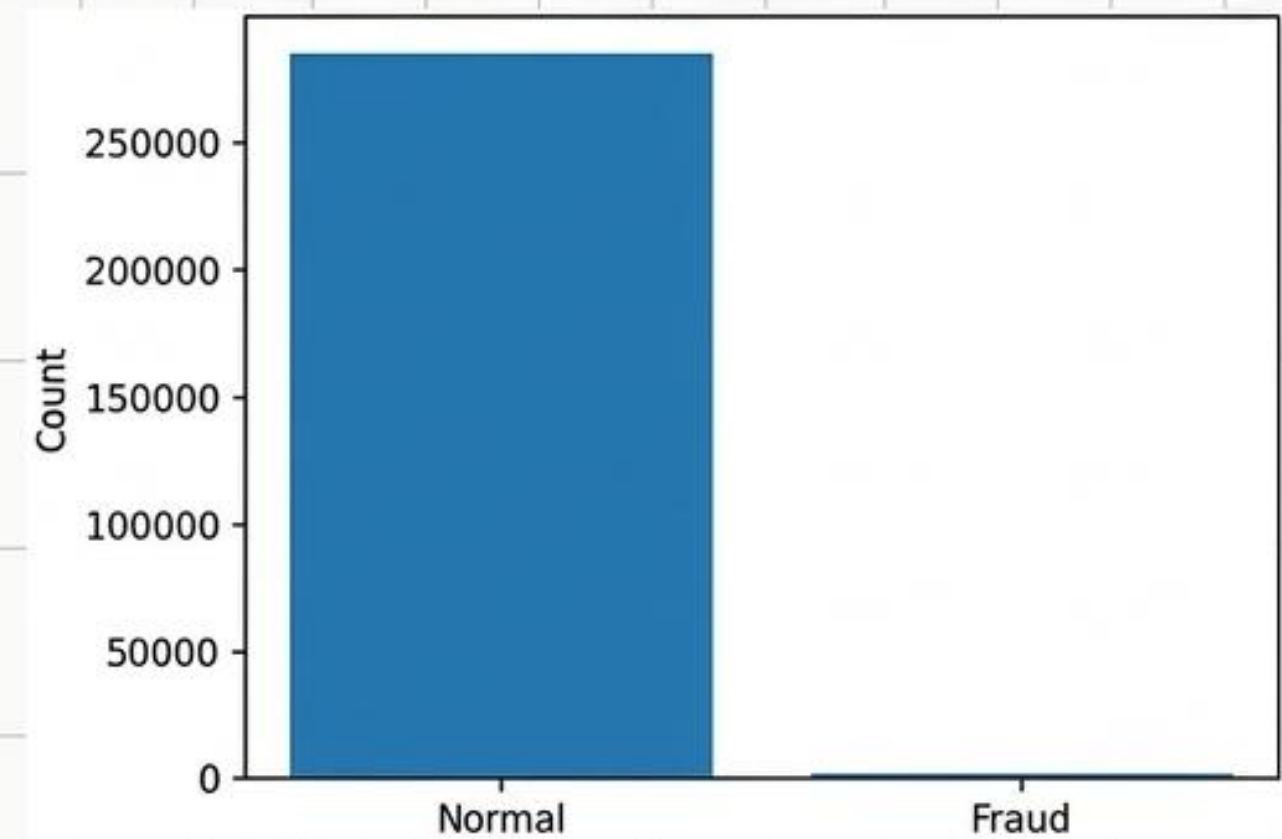
Course: DA5400 – Foundations of Machine Learning
Instructors: Prof. Nirav P Bhatt, Dr. P.S Jayadev

Class Imbalance in Critical Detection Systems

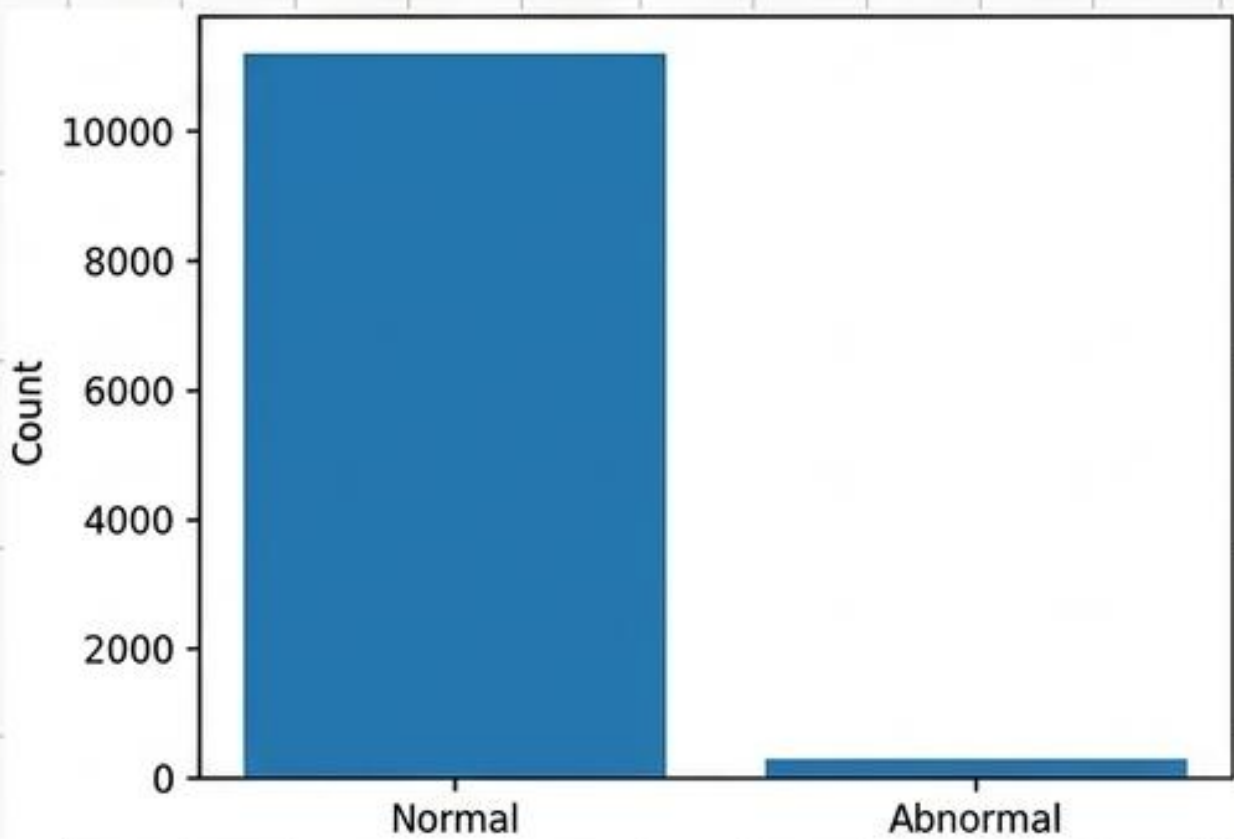
Traditional models inherently favor majority classes, causing failures on rare but critical events. Effectively handling class imbalance is vital in domains like fraud detection and medical diagnosis to prevent severe financial and medical consequences.

Dataset	Samples	Features	Minority Samples	Imbalance Ratio
Credit Card Fraud	284,807	30	492	577:1
Mammography	11,183	6	260	42:1

Credit Card Fraud: 577:1



Mammography: 42:1



Dynamic Learning via Adaptive Instance-Level Reweighting

Instance Difficulty Estimation

Measuring learning difficulty iteratively.

Adaptive Weighted Sampling

Favoring difficult samples in subsequent training.

Dynamic Sample Importance

Updating priorities without altering dataset distribution.

Adaptive Sampling Probability:

$$w_i = \frac{Difficulty_i}{\sum(Difficulty_j)}$$

Data Preprocessing and Normalization

- Baseline Models
- Logistic Regression
 - SVM
 - Random Forest

Performance Evaluation

Initial Prediction Probability Estimation

Instance Difficulty Estimation

Adaptive Weight Updating and Weighted Sampling

- Proposed Variants
- Adaptive Reweighting
 - Weighted-Loss Extension
 - Ensemble Extension

Performance Evaluation

Architecture & Models Spec Sheet

Neural Network: Input Feature -> FC (128) + ReLU + Dropout -> FC (64) + ReLU + Dropout -> FC (32) + ReLU -> Output (2)

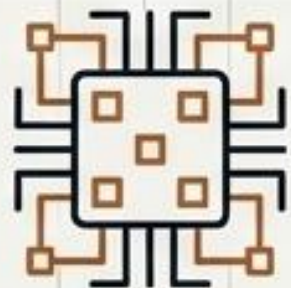
Baselines: Logistic Regression, SVM, Random Forest

Extensions: Weighted-Loss (Equation 5), Ensemble (Equation 6)

Experimental Setup and Technical Configuration



Framework
PyTorch



Hardware
CUDA GPU Acceleration

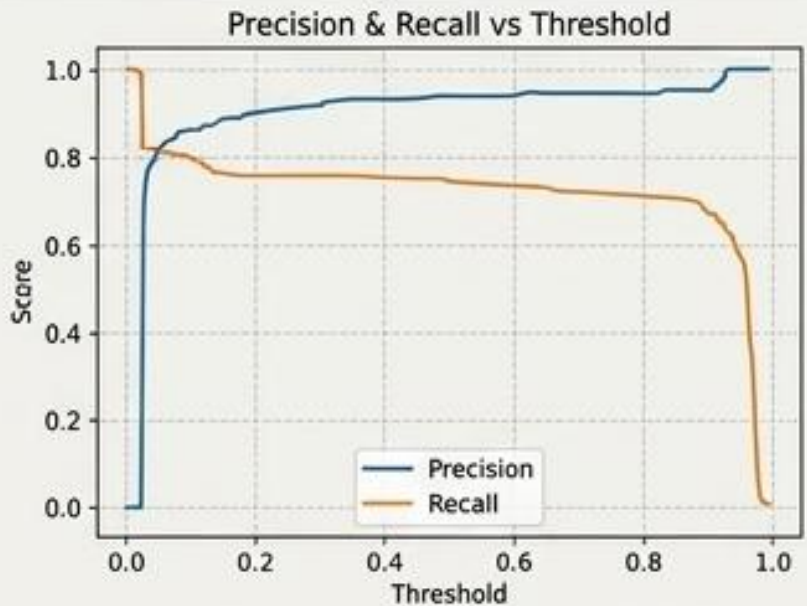
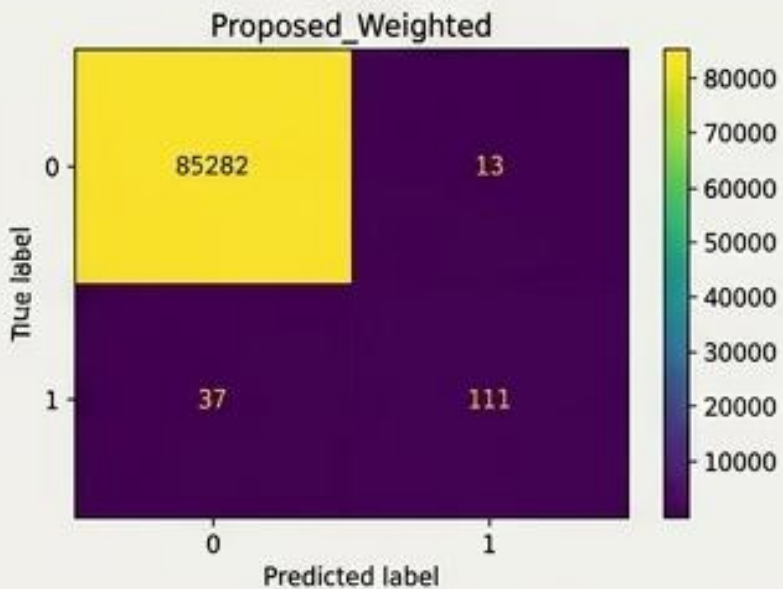
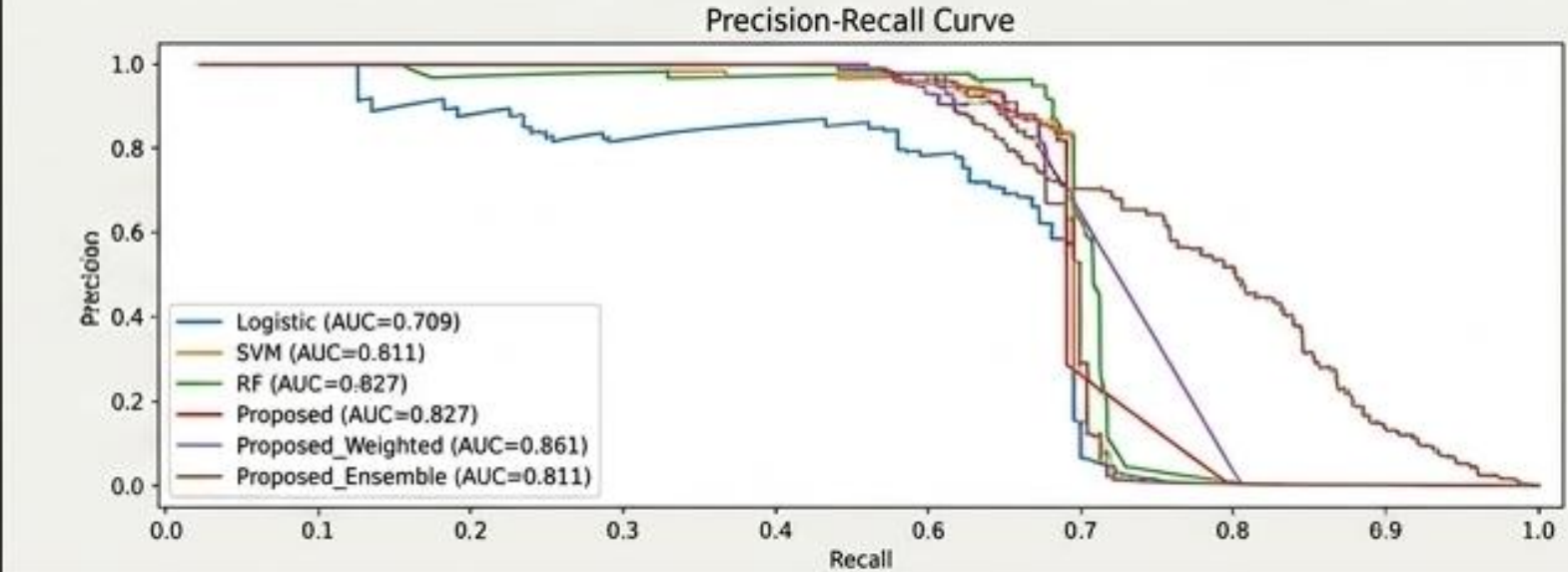
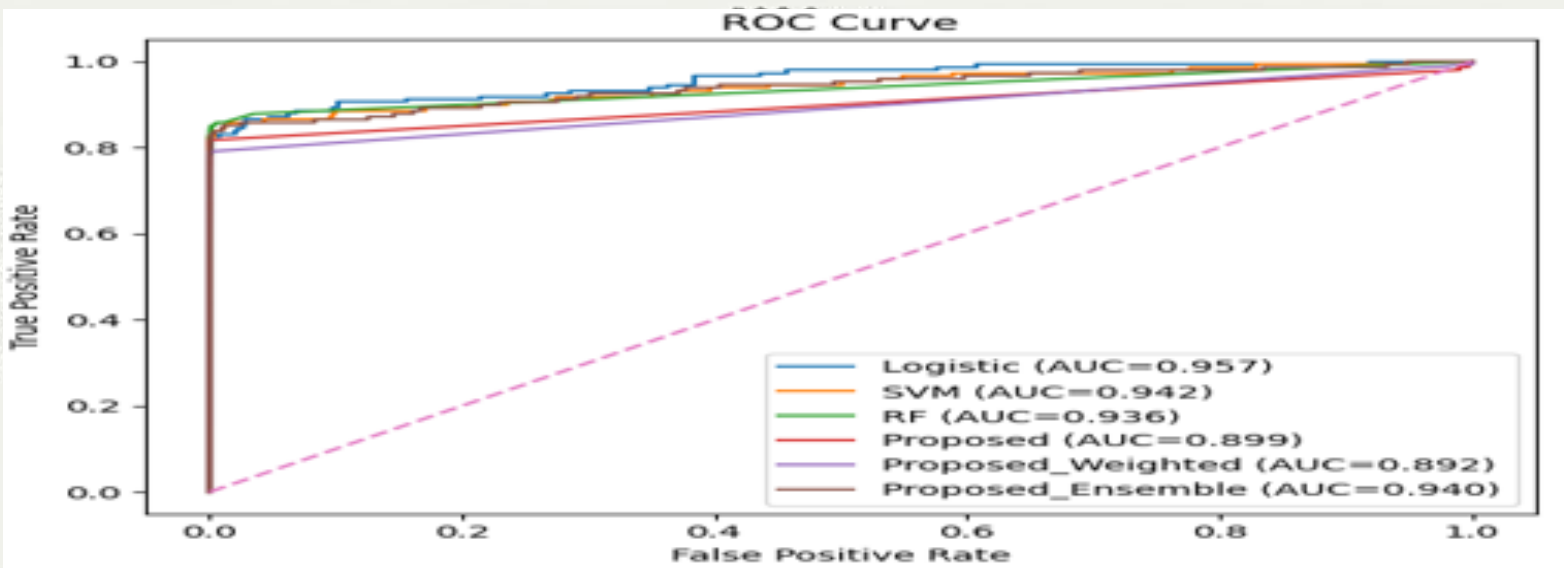


Optimization Strategy
Adam optimizer combined with weighted random sampling driven by dynamic difficulty scores.

Parameter	Value
Optimizer	Adam
Learning Rate	0.001
Batch Size	512
Epochs	100
Dropout Rate	0.3 / 0.2
Loss Function	Cross-Entropy Loss
Device	CUDA GPU

Extreme Imbalance (577:1): Credit Card Fraud Performance

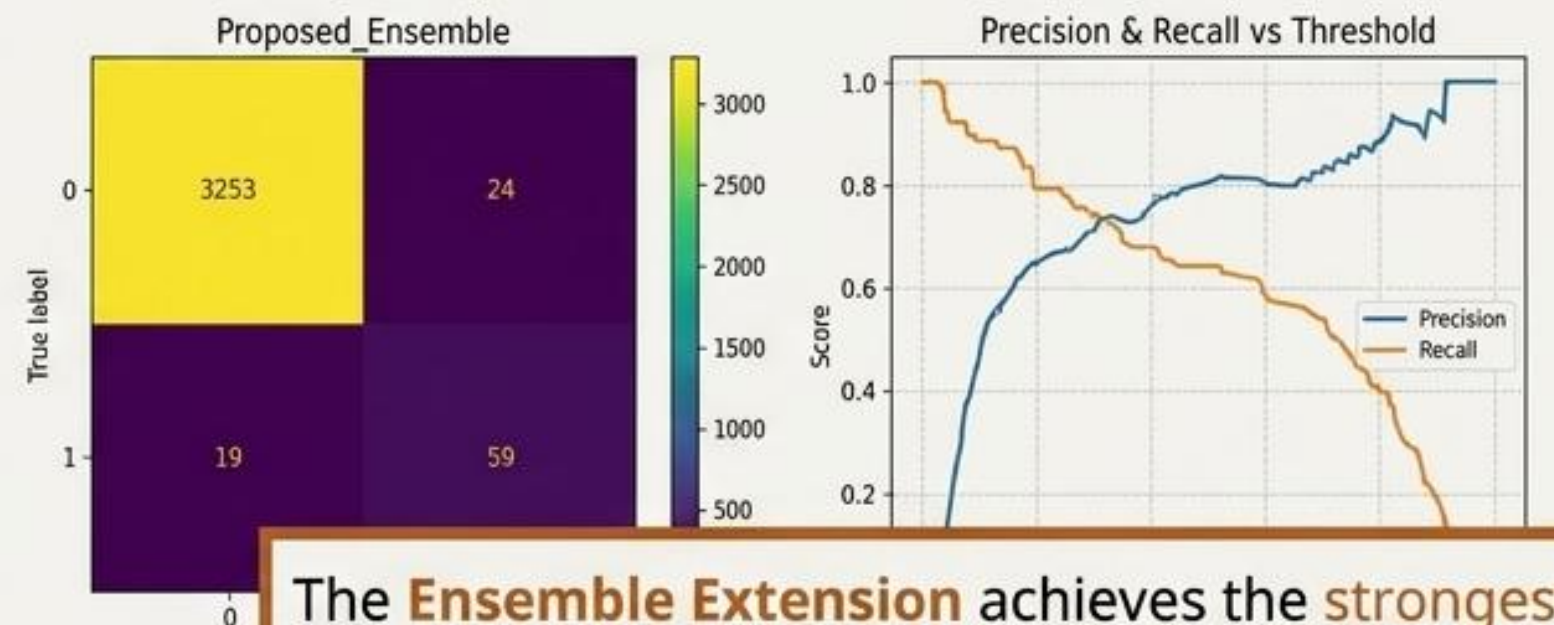
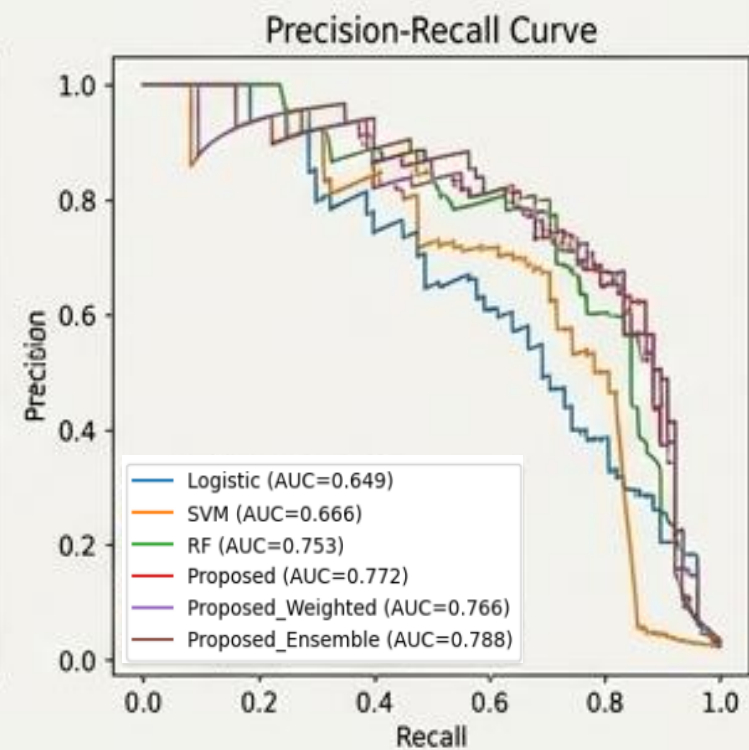
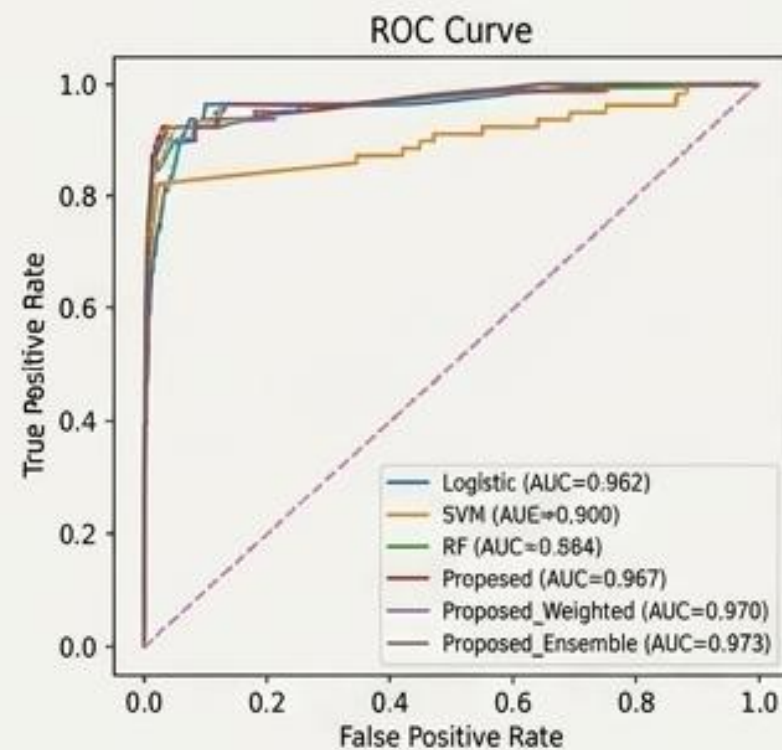
Method	Accuracy	Precision	Recall	F1-score	ROC-AUC	PR-AUC
Logistic Regression	0.99917	0.7895	0.7095	0.7473	0.9566	0.7088
SVM	0.99945	0.9244	0.7432	0.8240	0.9419	0.8114
Random Forest	0.99958	0.9667	0.7838	0.8657	0.9365	0.8274
Proposed Adaptive Reweighting	0.99949	0.9333	0.7568	0.8358	0.8994	0.8269
Weighted-Loss Extension	0.99947	0.9120	0.7703	0.8352	0.8918	0.8612
Ensemble Extension	0.99945	0.9244	0.7432	0.8240	0.9400	0.8111



The Weighted-Loss Extension maximizes PR-AUC (0.8612) and actively reduces false negatives, proving essential for extreme imbalance scenarios.

Moderate Imbalance (42:1): Mammography Performance

Method	Accuracy	Precision	Recall	F1-score	ROC-AUC	PR-AUC
Logistic Regression	0.9818	0.6024	0.6410	0.6211	0.9619	0.6490
SVM	0.9848	0.6667	0.6923	0.6792	0.8999	0.6660
Random Forest	0.9884	0.7671	0.7179	0.7417	0.9637	0.7530
Proposed Adaptive Reweighting	0.9875	0.7308	0.7308	0.7308	0.9668	0.7717
Weighted-Loss Extension	0.9878	0.7229	0.7692	0.7453	0.9696	0.7656
Ensemble Extension	0.9890	0.7971	0.7051	0.7483	0.9734	0.7884



The **Ensemble Extension** achieves the **strongest overall performance**, proving that prediction averaging maximizes robustness for moderately imbalanced datasets.

Theoretical Claims vs. Empirical Observations

Paper Claims	Experimental Verification
Adaptive instance-level weighting improves minority-class learning	✓ Improved Recall, F1-score, and PR-AUC compared to LR and SVM baselines
Difficult samples should receive higher importance during training	✓ Confusion matrices showed reduced false-negative predictions
Imbalance can be handled without oversampling or undersampling	✓ Competitive minority-class detection achieved without modifying dataset distribution
Different imbalance levels require different optimization strategies	✓ Weighted-Loss performed best on CC dataset, while Ensemble Extension achieved stronger performance on Mammography dataset

Project-Specific Conclusions

Pillar 1

1.

Adaptive Weighting
Effective

Adaptive instance-level weighting improved minority-class detection by emphasizing difficult training samples without oversampling or undersampling.

Pillar 2

2.

Imbalance Severity
Matters

Weighted-Loss performed best under extreme imbalance (CC dataset: 577:1), while Ensemble Extension achieved better robustness under moderate imbalance (Mammography dataset: 42:1).

Pillar 3

3.

Strong Random
Forest Baseline

Random Forest remained a highly competitive baseline model, especially for structured tabular datasets.

Thank You.